$E[XY] = E[(X - E(X) + E(X))(Y - E(Y) + E(Y))] = E[(X - E(X))(Y - E(Y))] + \underbrace{E[X - E(X)]E[Y]}_{=0} + \underbrace{E[X]E[Y - E(Y)]}_{=0} + E[X]E[Y] = E[X]E[Y] + Cov(X, Y)$

$E(X) = \pi_1 X_1 + \pi_2 X_2 + \cdots + \pi_n X_n$, $\sigma^2(X) = \pi_1[X_1 - E(X)]^2 + \pi_2[X_2 - E(X)]^2 + \cdots + \pi_n[X_n - E(X)]^2$, *covariance* between two random variables X_1 and X_2: $\sigma(X_1, X_2) = E\{[X_1 - E(X_1)][X_2 - E(X_2)]\} = E(X_1 Y_2) - E(X_1)E(X_2)$, *correlation* between X_1 and X_2: $-1 \leq \rho(X_1, X_2) = \frac{\sigma(X_1, X_2)}{\sigma(X_1)\sigma(X_2)} \leq 1$, 性质: *1.* $Cov(X_1, X_1) = Var(X_1)$, *2.* $Cov(aX_1, X_2) = aCov(X_1, X_2)$, *3.* $Cov(X_1 + c, X_2) = Cov(X_1, X_2)$, *4.* $Cov(X_1 + X_2, X_3) = Cov(X_1, X_3) + Cov(X_2, X_3)$, *5.* If $X_3 = X_1 + X_2$ then $Var(X_3) = Cov(X_3, X_3) = Cov(X_1 + X_2, X_1 + X_2) = Cov(X_1, X_1) + Cov(X_1, X_2) + Cov(X_2, X_1) + Cov(X_2, X_2) = Var(X_1) + Var(X_2) + 2Cov(X_1, X_2)$, *6.* $Var(aX_1 + bX_2) = a^2 Var(X_1) + b^2 Var(X_2) + 2abCov(X_1, X_2)$

$Y = \alpha + \beta X + \varepsilon$, $\beta = Cov(Y, X)/Var(X)$

The **line** originating at $(0, r_f)$ and running through $(\sigma_M, E(\tilde{r}_M))$ is called the **capital market line (CML)**. **All individually optimal portfolios** are located along the CML and are formed as combinations of the risk free asset and the market portfolio. The equation $E(\tilde{r}_P) = r_f + \left[\frac{E(\tilde{r}_M) - r_f}{\sigma_M}\right]\sigma_P$ implies that the market portfolio's Sharpe ratio $\frac{E(\tilde{r}_M) - r_f}{\sigma_M}$ measures the equilibrium **price of risk**: the expected return that each investor gives up when he or she adjusts his or her total portfolio to reduce risk.

CAPM: $E(\tilde{r}_j) = r_f + \beta_j[E(\tilde{r}_M) - r_f] \quad \beta_j = \frac{\sigma_{jM}}{\sigma_M^2}$ quadratic utility / normally-distributed returns: justify mean-variance utility

It implies that if we rank individual stocks or portfolios of stocks according to their betas, their expected returns should all lie along a single **security market line (SML)** with slope $E(\tilde{r}_M) - r_f$.

$E(\tilde{r}_j) = r_f + \left(\frac{\sigma_{jM}}{\sigma_M^2}\right)[E(\tilde{r}_M) - r_f] \overset{\rho_{jM} = \frac{\sigma_{jM}}{\sigma_j \sigma_M}}{=\!=\!=} E(\tilde{r}_j) = r_f + \left[\frac{E(\tilde{r}_M) - r_f}{\sigma_M}\right]\rho_{jM}\sigma_j$ The **idiosyncratic risk** in asset j, that is, the portion that is uncorrelated with the market return, can be diversified away by holding the market portfolio. Risk in asset j is priced only to the extent that it takes the form of **aggregate risk** that, because it is correlated with the market portfolio, cannot be diversified away. $\Longrightarrow$ *1.* Only assets with random returns that are positively correlated with the market return earn expected returns above the risk free rate. They must, in order to induce investors to take on more aggregate risk. *2.* Assets with returns that are uncorrelated with the market return have expected returns equal to the risk free rate, since their risk can be completely diversified away. *3.* Assets with negative betas – that is, with random returns that are negatively correlated with the market return – have expected returns **below** the risk free rate! For these assets, $E(\tilde{r}_j) - r_f < 0$ is like an "insurance premium" that investors will pay in order to insulate themselves from aggregate risk.

$P_t^C = \frac{E(\tilde{C}_{t+1})}{1 + r_f + \psi} \Big/ \frac{E(\tilde{C}_{t+1}) - \Psi}{1 + r_f} \quad 1 + \tilde{r}_C = \frac{\tilde{C}_{t+1}}{P_t^C} \Leftrightarrow E(\tilde{r}_C) = r_f + \beta_C[E(\tilde{r}_M) - r_f] \quad P_t^C = \frac{E(\tilde{C}_{t+1})}{1 + r_f + \beta_C[E(\tilde{r}_M) - r_f]} \Big/ \frac{E(\tilde{C}_{t+1}) - P_t^C \beta_C[E(\tilde{r}_M) - r_f]}{1 + r_f}$

APT: it applies only to "well-diversified portfolios" and "most" individual securities

The APT starts by assuming that asset returns are governed by a factor model such as the market model.

Market Model: $\tilde{r}_j = \alpha_j + \beta_j[\tilde{r}_M - E(\tilde{r}_M)] + \varepsilon_j \overset{\bar{\alpha}_j = \alpha_j - \beta_j E(\tilde{r}_M)}{=\!=\!=} \tilde{r}_j = \bar{\alpha}_j + \beta_j \tilde{r}_M + \varepsilon_j \overset{E(\tilde{r}_j) = \alpha_j}{=\!=\!=} E(\tilde{r}_j) + \beta_j[\tilde{r}_M - E(\tilde{r}_M)] + \varepsilon_j$

$\sigma_j^2 = E\{[\tilde{r}_j - E(\tilde{r}_j)]^2\} = E(\{\beta_j[\tilde{r}_M - E(\tilde{r}_M)] + \varepsilon_j\}^2) = \beta_j^2 E\{[\tilde{r}_M - E(\tilde{r}_M)]^2\} + 2\beta_j E\{[\tilde{r}_M - E(\tilde{r}_M)]\varepsilon_j\} + E(\varepsilon_j^2) = \beta_j^2\sigma_M^2 + 2\beta_j Cov(\tilde{r}_M, \varepsilon_j) + \sigma_{\varepsilon_j}^2 = \beta_j^2\sigma_M^2 + \sigma_{\varepsilon_j}^2$,

$Cov(\tilde{r}_j, \tilde{r}_k) = E\{[\tilde{r}_j - E(\tilde{r}_j)][\tilde{r}_k - E(\tilde{r}_k)]\} = E(\{\beta_j[\tilde{r}_M - E(\tilde{r}_M)] + \varepsilon_j\}\{\beta_k[\tilde{r}_M - E(\tilde{r}_M)] + \varepsilon_k\}) =$

$\beta_j\beta_k E\{[\tilde{r}_M - E(\tilde{r}_M)]^2\} + \beta_j E\{[\tilde{r}_M - E(\tilde{r}_M)]\varepsilon_k\} + \beta_k E\{[\tilde{r}_M - E(\tilde{r}_M)]\varepsilon_j\} + E(\varepsilon_j\varepsilon_k) = \beta_j\beta_k\sigma_M^2 + \beta_j Cov(\tilde{r}_M, \varepsilon_k) + \beta_k Cov(\tilde{r}_M, \varepsilon_j) + Cov(\varepsilon_j, \varepsilon_k) = \beta_j\beta_k\sigma_M^2$, allows these variances and covariances to be estimated based on only $2J + 1$ underlying "parameters:" the J β_j's, the J $\sigma_{\varepsilon_j}^2$'s, and the 1 σ_M^2.

There are enough individual assets to create many **well-diversified** portfolios. $E(\tilde{r}_w) = \sum_{i=1}^N w_i E(\tilde{r}_i) = \left(\frac{1}{N}\right)\sum_{i=1}^N E(\tilde{r}_i)$, $\beta_w = \sum_{i=1}^N w_i\beta_i = \left(\frac{1}{N}\right)\sum_{i=1}^N \beta_i$, $\varepsilon_w = \sum_{i=1}^N w_i\varepsilon_i = \left(\frac{1}{N}\right)\sum_{i=1}^N \varepsilon_i$, $\sigma_{\varepsilon_w}^2 = \sum_{i=1}^N w_i^2 E(\varepsilon_i^2) + \sum_{i=1}^N \sum_{h \neq i} w_i w_h E(\varepsilon_i \varepsilon_h) = \sum_{i=1}^N w_i^2 E(\varepsilon_i^2) = \sum_{i=1}^N w_i^2 \sigma_{\varepsilon_i}^2$, one in which the number of assets included is sufficiently large to make the portfolio's idiosyncratic risk vanish. The return on any portfolio **well-diversified** portfolio is $\tilde{r}_w = E(\tilde{r}_w) + \beta_w[\tilde{r}_M - E(\tilde{r}_M)]$

Investors act to eliminate arbitrage opportunities across all well-diversified portfolios.

Propositions: *1.* In the absence of arbitrage opportunities, well-diversified portfolios with the same betas **must** have the same expected returns. *2.* The absence of arbitrage opportunities requires the expected returns on all well-diversified portfolios to satisfy $E(\tilde{r}_w) = r_f + \beta_w[E(\tilde{r}_M) - r_f]$ *Proof* of APT linear pricing: *1.* Benchmark Portfolios: Port 1 is well-diversified with $\beta = 0$ (synthetic risk-free), so $\tilde{r}_w^1 = E(\tilde{r}_w^1) = r_f$; Port 2 is well-diversified with $\beta = 1$ (market portfolio), so $\tilde{r}_w^2 = \tilde{r}_M$ and $E(\tilde{r}_w^2) = E(\tilde{r}_M)$. *2.* Assumption for Contradiction: Assume a well-diversified Port 3 with β_w has $E(\tilde{r}_w^3) = r_f + \beta_w[E(\tilde{r}_M) - r_f] + \Delta$ ($\Delta \neq 0$), making its return $\tilde{r}_w^3 = r_f + \beta_w[E(\tilde{r}_M) - r_f] + \Delta + \beta_w[\tilde{r}_M - E(\tilde{r}_M)]$. *3.* Replicating Portfolio: Form Port 4 with weight $1 - \beta_w$ in Port 1 and β_w in Port 2, yielding return $\tilde{r}_w^4 = (1 - \beta_w)r_f + \beta_w\tilde{r}_M = r_f + \beta_w[E(\tilde{r}_M) - r_f] + \beta_w[\tilde{r}_M - E(\tilde{r}_M)]$. *4.* Arbitrage Strategy: Ports 3 and 4 have identical betas. If $\Delta > 0$, long Port 3 and short Port 4; if $\Delta < 0$, long Port 4 and short Port 3. Both strategies are zero net investment (self-financing) at $t = 0$ but yield a guaranteed riskless profit $|\Delta| > 0$ at $t = 1$. *5.* Conclusion: This violates the no-arbitrage assumption, thus $\Delta = 0$ must hold, proving all well-diversified portfolios strictly satisfy $E(\tilde{r}_w) = r_f + \beta_w[E(\tilde{r}_M) - r_f]$.

Investor j chooses c_j^0 and c_j^i for all $i = 1, 2, \ldots, N$ to maximize $u_j(c_j^0) + \beta \sum_{i=1}^N \pi_i u_j(c_j^i)$ s. t.

$w_j^0 + \sum_{i=1}^N q^i w_j^i \geq c_j^0 + \sum_{i=1}^N q^i c_j^i$, $u_j(c_j^0) + \beta \sum_{i=1}^N \pi_i u_j(c_j^i) + \lambda_j \left(w_j^0 + \sum_{i=1}^N q^i w_j^i - c_j^0 - \sum_{i=1}^N q^i c_j^i \right) \implies u_j'(c_j^0) - \lambda_j = 0$,

$\beta \pi_i u_j'(c_j^i) - \lambda_j q^i = 0 \quad \forall i = 1, 2, \ldots, N \implies q^i = \frac{\beta \pi_i u_j'(c_j^i)}{u_j'(c_j^0)} \; \forall i = 1, 2, \ldots, N$ q^i is higher if investor j's consumption falls between $t = 0$ and state i at $t = 1$ $\implies$ During a recession

1. The resource allocation from a competitive equilibrium is Pareto optimal, 2. Any Pareto optimal resource allocation can be supported in a competitive equilibrium.

Optimal Risk Sharing(Social Planner): Max weighted expected utility $\max \theta\{u_1(c_1^0) + \beta[\pi_1 u_1(c_1^1) + \pi_2 u_1(c_1^2)]\} + (1-\theta)\{u_2(c_2^0) + \beta[\pi_1 u_2(c_2^1) + \pi_2 u_2(c_2^2)]\}$ s.t. aggregate resource constraints $w^i = c_1^i + c_2^i (i = 0, 1, 2)$. Lagrangian FOC yields core risk-sharing condition: $\theta \beta \pi_i u_1'(c_1^i) = (1-\theta)\beta \pi_i u_2'(c_2^i)$, simplifying to $\theta u_1'(c_1^i) = (1-\theta) u_2'(w^i - c_1^i)$. *Case 1(No Aggregate Risk)*: If $w^1 = w^2$, then $c_1^1 = c_1^2$ and $c_2^1 = c_2^2$; agents perfectly diversify idiosyncratic risk. *Case 2(Aggregate Risk + Identical CRRA* $u_j(c) = \frac{c^{1-\gamma}-1}{1-\gamma}$ *and* $u'(c) = c^{-\gamma}$*)*: Substituting gives $\theta (c_1^i)^{-\gamma} = (1-\theta)(w^i - c_1^i)^{-\gamma}$, solving for fixed proportional shares: $c_1^i = [\theta^{1/\gamma}/(\theta^{1/\gamma} + (1-\theta)^{1/\gamma})] w^i$ and $c_2^i = [(1-\theta)^{1/\gamma}/(\theta^{1/\gamma} + (1-\theta)^{1/\gamma})] w^i$. *Special Case(Log Utility* $\gamma = 1$*)*: Consumption shares exactly equal planner weights: $c_1^i = \theta w^i$, $c_2^i = (1-\theta) w^i$. *Conclusion*: A-D model completely diversifies idiosyncratic risk; aggregate risk is shared proportionally across agents.

Euler Equations & No-Arbitrage Pricing: An asset paying X_i in state i is replicated by contingent claims, making its price $P^A = \sum_{i=1}^N q^i X_i$. Substituting the A-D equilibrium state price $q^i = \frac{\beta \pi_i u_j'(c_j^i)}{u_j'(c_j^0)}$ gives the price Euler equation $P_A = \sum_{i=1}^N \left[\frac{\beta \pi_i u_j'(c_j^i)}{u_j'(c_j^0)} \right] X_i$, or equivalently $u_j'(c_j^0) P_A = \beta E[u_j'(\tilde{c}_j) \tilde{X}]$. By defining asset return as $R_i = \frac{X_i}{P^A}$, obtain the return Euler equation $1 = \beta E \left[\frac{u_j'(\tilde{c}_j)}{u_j'(c_j^0)} \tilde{R} \right]$. Defining the random **Intertemporal Marginal Rate of Substitution (IMRS)** as $\tilde{m}_j = \frac{\beta u_j'(\tilde{c}_j)}{u_j'(c_j^0)}$, the equations condense into the fundamental asset pricing formula: $1 = E(\tilde{m}_j \tilde{R})$. Without uncertainty, this reduces to Fisher's interest rate theory: $\frac{1}{1+r} = \frac{\beta u_j'(c_j^1)}{u_j'(c_j^0)}$.

Financial markets are **complete** if, for each possible state of each future date, there exists a market for a contingent claim that pays off one unit of consumption in that date-state combination and zero otherwise.

Infer contingent claim prices q^i from complex assets. **Method:** For each state i, solve a linear system to find portfolio weights $w = [w^1, w^2, w^3]$ that yield payoff 1 in state i and 0 otherwise, then compute $q^i = \sum w^j p_j$. *State 1 (*q^1*)*: Solve $w_1^1 z_1^1 + w_1^2 z_2^1 + w_1^3 z_3^1 = 3w_1^1 + w_1^2 + 2w_1^3 = 1$, $w_1^1 z_1^2 + w_1^2 z_2^2 + w_1^3 z_3^2 = 2w_1^1 + w_1^2 = 0$, $w_1^2 + 2w_1^3 = 0 \Rightarrow$ $w = [1/3, -2/3, 1/3] \Rightarrow q^1 = (1/3)(2.20) - (2/3)(0.95) + (1/3)(1.50) = 0.60$

Let $C_1^i(S, 2)$ denote the payoffs generated by the call option on the stock S with strike price $K = 2$ with expiration date at $T = 1$ in each state $i = 1, 2, 3$.

Case 1: Call always ITM. Payoffs are $C^G = P^G - K$ and $C^B = P^B - K$. Replicate with S shares and B bonds: $SP^G + B(1 + r_f) = P^G - K$ and $SP^B + B(1 + r_f) = P^B - K$. Solving yields $S = 1$ and $B = -\frac{K}{1+r_f}$. By no-arbitrage, initial option price V_0 equals initial portfolio cost: $V_0 = P_0 - \frac{K}{1+r_f}$. Expressed in general form $V_0 = N_1 P_0 - N_2 \frac{K}{1+r_f}$, we have $N_1 = 1, N_2 = 1$.

Case 2: Call ITM in Good State only. Payoffs: $C^G = P^G - K$, $C^B = 0$. Replicate: $SP^G + B(1 + r_f) = P^G - K$ and $SP^B + B(1 + r_f) = 0$. Solve: $S = \frac{P^G - K}{P^G - P^B}$ and $B = -\frac{P^B(P^G - K)}{(1+r_f)(P^G - P^B)}$. No-arbitrage price $V_0 = \frac{P^G - K}{P^G - P^B} P_0 - \frac{P^B \left(\frac{P^G}{P^G} - 1 \right)}{P^G - P^B} \frac{K}{1+r_f}$. In general form $V_0 = N_1 P_0 - N_2 \frac{K}{1+r_f}$, we get $N_1 = \frac{P^G - K}{P^G - P^B}$ and $N_2 = \frac{P^B \left(\frac{P^G}{P^G} - 1 \right)}{P^G - P^B}$, with $0 < N_1, N_2 < 1$.

Case 3: Call OTM in both. Payoffs $C^G = 0, C^B = 0$. Replicated by 0 shares and 0 bonds, so $V_0 = 0$, meaning $N_1 = 0, N_2 = 0$.

Basic: $1 = E[m R_i]$, SDF $m = \beta u'(c_j^i)/u'(c_j^0)$. Risk-free: $E[m] = 1/R_f$. Risky k: $1 = E[m]E[R^k] + Cov(m, R^k) \Rightarrow$ $E[R^k] - R_f = -R_f Cov(m, R^k)$. Intuition: m moves opposite to c. If R^k rises with c (no insurance), $Cov(m, R^k) < 0 \Rightarrow$ higher $E[R^k]$. If R^k falls with c (insurance), lower $E[R^k]$. Market m: $E[R^m] - R_f = -R_f Cov(m, R^m)$. Ratio (1)/(2): $[E(R^k) - R_f]/[E(R^m) - R_f] = Cov(m, R^k)/Cov(m, R^m)$. If $m = R^m + const$, ratio becomes $\beta \Rightarrow$ CAPM: $E[R^k] - R_f = \beta(E[R^m] - R_f)$. Difference: CAPM risk comoves with market return ($m = R^m + c$); AD risk comoves with aggregate consumption ($m = \beta u'(c_j^i)/u'(c_j^0)$).